

HENDRICKSON CREEK, AB, Canada

WMO: 719790

Lat: 53.795N

Lon: 118.447W

Elev: 4751

StdP: 12.34

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.0	-23.6	-38.3	0.7	-31.7	-29.4	1.3	-23.3	14.4	31.2	13.6	31.0	1.5	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	77.1	55.4	73.4	54.1	70.1	52.8	57.8	71.1	56.1	69.3	54.5	67.1	5.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.2	71.8	60.7	51.1	66.4	58.5	49.1	61.6	56.9	27.0	71.5	25.8	69.3	24.7	67.5	64.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.6	9.5	DB	-42.1	82.7	5.3	3.3	-46.0	85.0	-49.1	87.0	-52.1	88.8	-56.0	91.2	(4)
(5)			WB	-40.3	61.1	7.6	1.8	-45.7	62.4	-50.2	63.4	-54.5	64.4	-60.0	65.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	13.4	15.4	20.9	30.8	40.6	47.5	52.3	50.8	43.6	33.1	19.6	11.9		
	DBStd	17.94	16.41	13.99	12.74	8.39	6.96	5.08	5.18	5.70	7.53	8.80	13.11	14.32		
	HDD50	6923	1136	969	903	577	299	107	35	61	217	525	912	1182		
	HDD65	12142	1601	1389	1368	1027	758	525	393	440	643	989	1362	1647		
	CDD50	256	0	0	0	0	6	31	107	86	25	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	277	0	0	0	0	6	29	112	92	37	1	0	0		
	CDH80	31	0	0	0	0	0	4	12	11	4	0	0	0		
	WSAvg	3.7	3.6	3.5	3.6	4.0	4.1	4.4	3.7	3.7	3.3	4.0	3.0	3.7		
	PrecAvg	25.3	1.7	1.1	1.5	1.3	2.3	3.5	3.5	2.8	2.0	1.9	2.0	1.7		
(15) Precipitation	PrecMax	31.7	3.4	2.1	2.5	2.5	4.3	5.6	7.0	4.3	3.9	3.1	4.0	2.9		
	PrecMin	19.4	0.4	0.4	0.4	0.6	1.2	1.1	1.7	1.4	0.7	1.0	0.7	0.8		
	PrecStd	2.7	0.7	0.4	0.5	0.5	0.7	1.1	1.3	0.8	0.7	0.6	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.6	48.5	55.3	64.4	73.8	78.3	81.5	81.5	78.8	67.5	51.4	45.3		
		MCWB	36.6	36.0	39.8	44.3	49.8	54.9	56.3	56.4	53.7	48.0	40.6	36.5		
	2%	DB	42.4	43.3	47.5	56.4	68.2	71.1	77.0	76.1	72.7	58.6	44.3	40.6		
		MCWB	34.3	33.7	36.2	41.6	48.0	52.0	55.6	56.0	52.8	45.1	36.2	33.0		
	5%	DB	39.0	38.8	43.2	51.2	63.7	67.4	73.3	72.0	67.6	52.8	39.9	36.0		
		MCWB	32.3	31.6	33.6	38.6	46.5	50.1	54.7	54.6	50.8	42.1	33.7	30.2		
	10%	DB	34.8	34.9	39.5	46.8	58.8	63.3	69.5	67.8	61.4	47.6	36.2	31.6		
		MCWB	29.9	29.1	31.2	35.9	44.3	48.7	53.7	53.5	48.9	39.3	31.2	27.5		
	0.4%	WB	37.3	37.7	40.6	45.8	51.6	56.6	60.4	59.9	57.0	49.8	40.9	37.1		
		MCDB	45.8	47.1	54.5	62.3	67.6	70.8	71.2	71.7	71.3	65.0	50.5	44.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	34.2	36.8	42.3	49.3	54.0	58.0	57.9	54.2	45.7	37.4	33.4		
		MCDB	41.6	41.5	46.1	54.6	64.8	67.5	71.5	70.8	69.8	56.2	43.4	39.6		
	5%	WB	32.7	31.9	34.1	39.3	47.3	51.9	56.2	56.3	51.8	42.9	34.3	30.7		
		MCDB	38.0	38.6	42.2	50.1	61.3	63.9	69.6	68.6	66.4	52.1	39.0	35.5		
	10%	WB	30.2	29.2	31.8	36.6	45.2	50.1	54.6	54.4	49.2	39.7	31.8	27.5		
		MCDB	34.9	34.6	38.5	45.3	57.3	60.5	67.4	65.8	60.5	46.7	35.2	31.5		
	5% DB	MDBR	23.6	26.2	26.9	24.9	25.9	25.6	27.5	27.9	26.9	22.8	22.1	23.4		
		MCDBR	23.7	29.0	30.5	32.7	38.7	36.8	38.5	38.4	40.3	32.6	25.0	25.6		
		MCWBR	16.8	20.5	20.0	18.9	20.1	18.5	19.2	20.0	21.6	19.4	17.7	19.0		
		MCWBR	21.6	28.0	28.4	30.5	35.3	31.2	34.1	32.9	37.3	30.0	23.1	24.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	15.8	20.0	18.9	18.0	19.2	16.8	18.0	18.4	20.5	18.3	17.2	18.8		
	Clear-Sky Solar Irradiance	taub	0.233	0.239	0.272	0.301	0.323	0.331	0.323	0.324	0.293	0.270	0.239	0.225		
		taud	2.461	2.530	2.462	2.420	2.431	2.443	2.505	2.489	2.590	2.611	2.518	2.395		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	252	287	295	297	293	291	292	285	284	271	247	229		
		Edh at Noon	15	20	27	32	34	34	31	30	23	18	14	13		
(44) All-Sky Solar Radiation	RadStd		249	522	893	1359	1580	1665	1707	1440	1007	562	270	192		
			20	49	68	95	131	142	136	97	115	44	24	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page